Institutional Partners

DisSModel

Building Open & Reproducible Geospatial Simulations with Python

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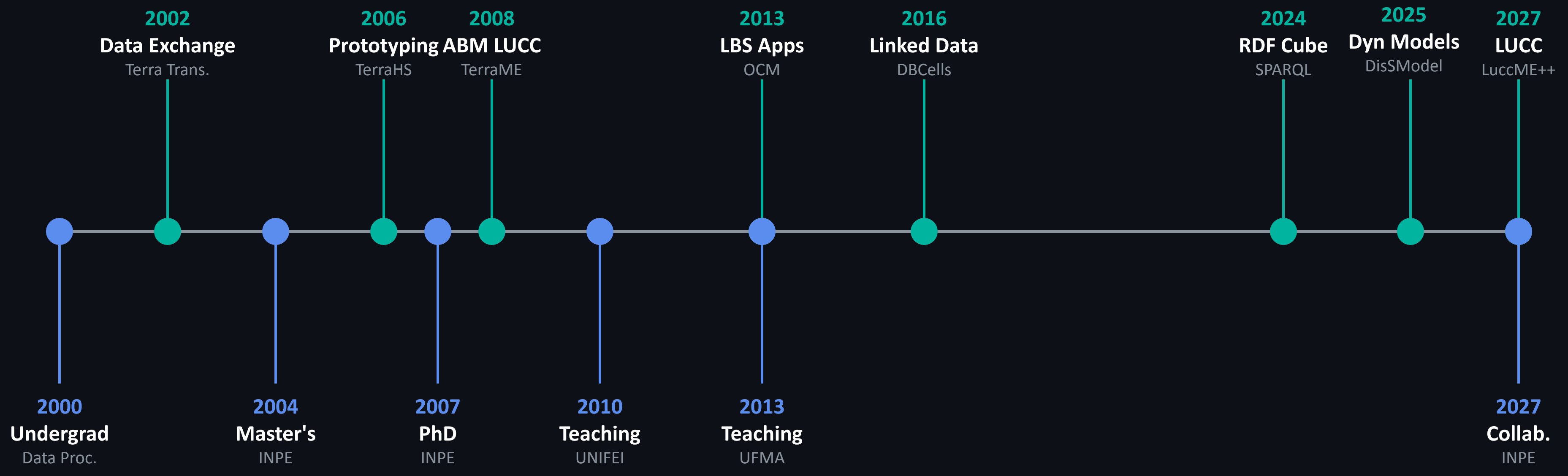
UFMA / LambdaGeo

Career & Research

Focus on Spatial Modeling and Geographic Data — Open Science



Research Projects

Academic & Professional

Science is not reproducible.

This is a documented crisis — not an opinion.

70%

of researchers cannot reproduce
experiments from other groups.

Baker (2016) · Nature

Three Root Failures

Code doesn't run

Implicit environments — works only on author's machine.

Data is missing

Input data not versioned. No data = no audit.

Parameters unknown

Configs live in ad hoc scripts or researcher's memory.

We model how territory changes.

Deforestation

Urban Expansion

Coastal Dynamics

Agricultural Frontier

Focus: ***Process-based geospatial modeling*** *— not general-purpose ML.*

Existing tools are powerful...

...but they weren't built for reproducibility at scale.

Dinamica EGO	No support for process-based geospatial models (INPE)
TerraME	Last release 2020 · No cloud/STAC support
LuccME	Inherits Lua constraints · Weak Python link

DisSModel

Discrete Spatial Modeling Framework for Python

Open source · Reproducible · Cloud-native

```
pip install dissmodel
```

"Science should not need to be rewritten to go into production."

— DisSModel Principle

End-to-End Workflow

1

Create

Design model using base library

2

Package

Encapsulate in executors

3

Publish

Push to GitHub + catalog

4

Run

Same code: local or cloud

Core Pillars

Reproducibility

Executor + TOML
Docker isolation
SHA-256 audit trail

FAIR Software

Findable via Zenodo DOI
Accessible via REST API
Interoperable + Reusable

Layer Separation

Science: pure logic
Platform: APIs & infra
SimOps: operations

Three Components

Library
`dissmodel`

- Core simulation engine.
- Local dev & execution.
- `pip install dissmodel`

Platform
`dissmodel-platform`

- Cloud-native infra.
- FastAPI + Docker + Redis.
- `docker compose up`

Catalog
`dissmodel-configs`

- Single source of truth.
- TOML manifests.
- Version-controlled PRs

Same code. Runs locally and on the remote platform.

Validation through laboratory models.

Cellular Automata · System Dynamics

Game of Life

DisSModel

Cellular Automaton · Built on GeoDataFrames

game_of_life.py

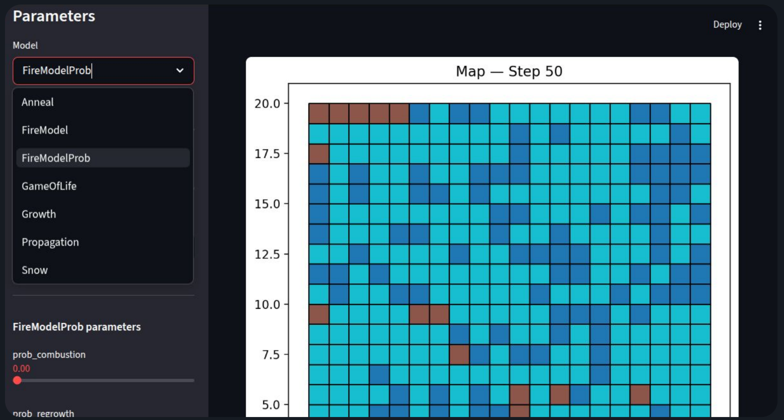
```python
class GameOfLife(CellularAutomaton):

    def rule(self, idx) -> int:
        state = self.gdf.loc[idx, 'state']
        n = self.neighbor_values(idx).sum()

        if state == 1:
        return 1 if 2 <= n <= 3 else 0

        return 1 if n == 3 else 0

# Environment Initialization
gdf = vector_grid((20, 20), attrs={"state": 0})
env = Environment()
GameOfLife(gdf).initialize()
```



Cellular Automata Simulation

Visualization of grid state at step 50. The model uses a **Queen** neighborhood (8 neighbors) and classic Conway's rules.

Explore the full code at **github.com/DisSModel**

SIR Model

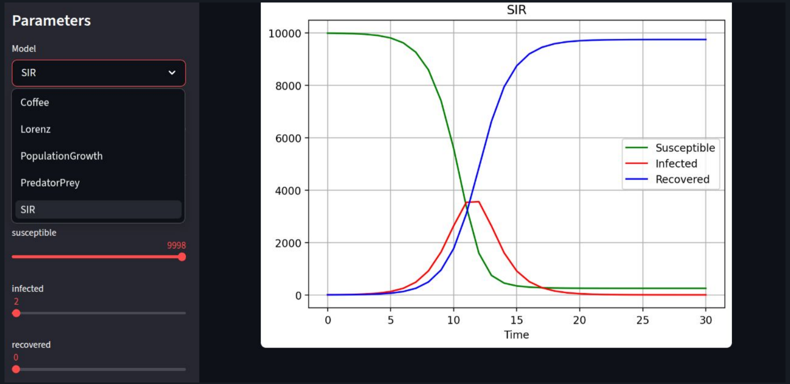
System Dynamics · Flows, Stocks, and Feedback Loops

sir_model.py

```python
class SIR:
  def execute(self):
    total = S + I + R
    alpha = contacts * prob
    new_inf = I * alpha * (S/total)
    new_rec = I / duration

    S -= new_inf
    I += new_inf - new_rec
    R += new_rec

env = Environment()
SIR(9998, 2, 0, 2, 6, 0.25)
Chart(title="SIR Model")
env.run(30)
```



Dynamic Simulation Output

Visualization of Susceptible (green), Infected (red), and Recovered (blue) populations over 30 time steps. Parameters control the interaction rates and duration.

Explore the full code at **github.com/DisSModel**

Now: real-world empirical models.

From theory to application.

brmangue

Coastal dynamics model · Maranhão Coast, Brazil

🌊 FloodModel

Simulates flooding due to sea level rise scenarios based on terrain elevation and elevation rate (mm/yr).

📍 Study Area

Ilha do Maranhão · Vector spatial grid · 97,309 cells
Scientific basis: Bezerra et al. (2014)

🌿 MangroveModel

Models mangrove migration driven by tidal height and soil type dynamics over time.

⚙️ Key Parameters

Elevation rate · Tide height · Active accretion
Simulation: 2008 → 2030 (88 steps)
CRS: EPSG:31984 · Resolution: 100m

Coastal Case Study: brmangue

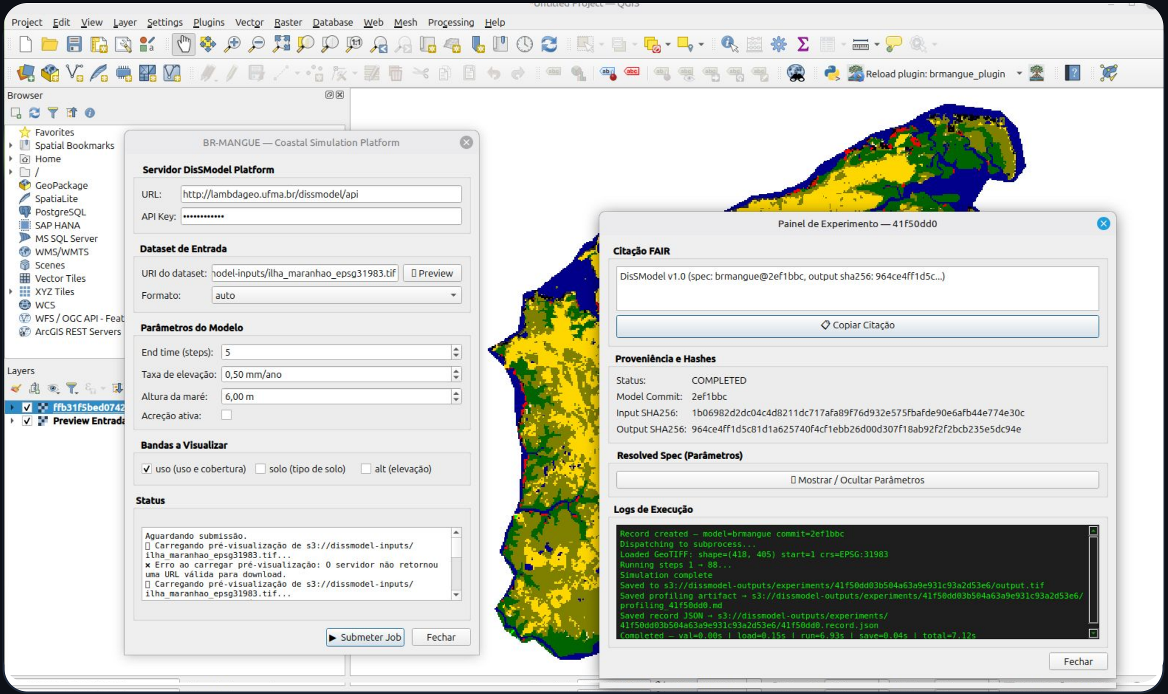
Mangrove migration + flooding · Maranhão Coast, Brazil

39.4×	99.8%	97K
Raster speedup vs Vector	Accuracy match vs TerraME	cells at scale

QGIS Plugin available — simulation inside the GIS environment, no API calls required.

Coastal Case Study: brmangue

Plugin Integration & Visual Interface



Plugin Features

Integrates `brmangue` model for coastal dynamics simulation and visualization.

- Intuitive parameter input
- Direct output visualization
- Seamless vector grid integration

QGIS Plugin interface: Simulation executed directly within the GIS environment.

DisSLUCC

Land Use & Cover Change — Continuous CLUE-type allocation

1 Demand

Magnitude of change per timestep. Pre-computed via historical CSV data (MapBiomas/LuccME).

```
DemandPreComputedValues(
  annual_demand=load_csv(...)
)
```

2 Potential

Change suitability via linear regression with spatial driver variables.

```
PotentialLinearRegression(
  gdf=gdf, drivers=[...]
)
```

3 Allocation

CLUE-like distribution: cells receive fractional proportions, not discrete classes.

```
AllocationClueLike(
  demand, potential
)
```

Veldkamp & Fresco (1996) CLUE · Evaluation: MAE (Willmott 2005) · Dataset: LuccME AC 2008–2014 · 6,574 cells

DisSLUCC — Land Use & Cover Change

CLUE-type allocation · LuccME AC dataset (2008–2014)

🚀 Performance Speedup

3.9×

Raster speedup vs Vector

31.6 ms vs 122.5 ms per step

🎯 Accuracy Metrics

MAE = 0.002276

Accuracy vs TerraME

- Vector = Raster on both substrates
- Preserves full algorithmic correctness

Evaluation based on Willmott (2005) methodology

What Do We Gain?

🔧 Hidden Engineering

Data orchestration runs seamlessly — infrastructure is invisible to the researcher.

🔍 Full Traceability

SHA-256 hashes + FAIR citations guarantee provenance at every simulation stage.

🗺️ Direct GIS Delivery

Results rendered natively in QGIS — no middleware, no extra steps.

🎯 End-to-End Focus

Researchers stay focused on science. Users stay focused on decisions.

Roadmap 2026–2028

2026 S1

JOSS Publication

PyPI v0.5
Formalisation at UFMA

2026 S2

Project Kickoff

DisSModel v1.0
CLUE & CLUE-S

2027 S1

BDC Integration

STAC reader
Xarray (time,y,x)

2027 S2

Dask / Ensemble

Parallel scenarios
Native Zarr

2028

Book + Community

PT-BR book
INPE workshop

Technology solves the how.

Governance solves the who and the where.

Infrastructure

Who maintains the platform? Server, storage, uptime.

Data Policy

Where do results live? Integration with Brazil Data Cube?

Model Catalog

Who reviews PRs? Acceptance criteria?

Community

Without shared infra, reproducibility problems remain.

DisSModel → LuccME++

DisSModel
Innovation Lab

LuccME++
Future of LUCC

2027/28 — DisSModel as the innovation laboratory for LuccME at CCST/INPE.

TerraME was the starting point — DisSModel advances as the foundation.

Reproducibility is not optional.

It is the foundation of science.

Thank you!

DisSModel — open source · open science

GitHub

`dissmodel.github.io`

Docs

`dissmodel.github.io/dissmodel/`

Install

`pip install dissmodel`

Contact

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